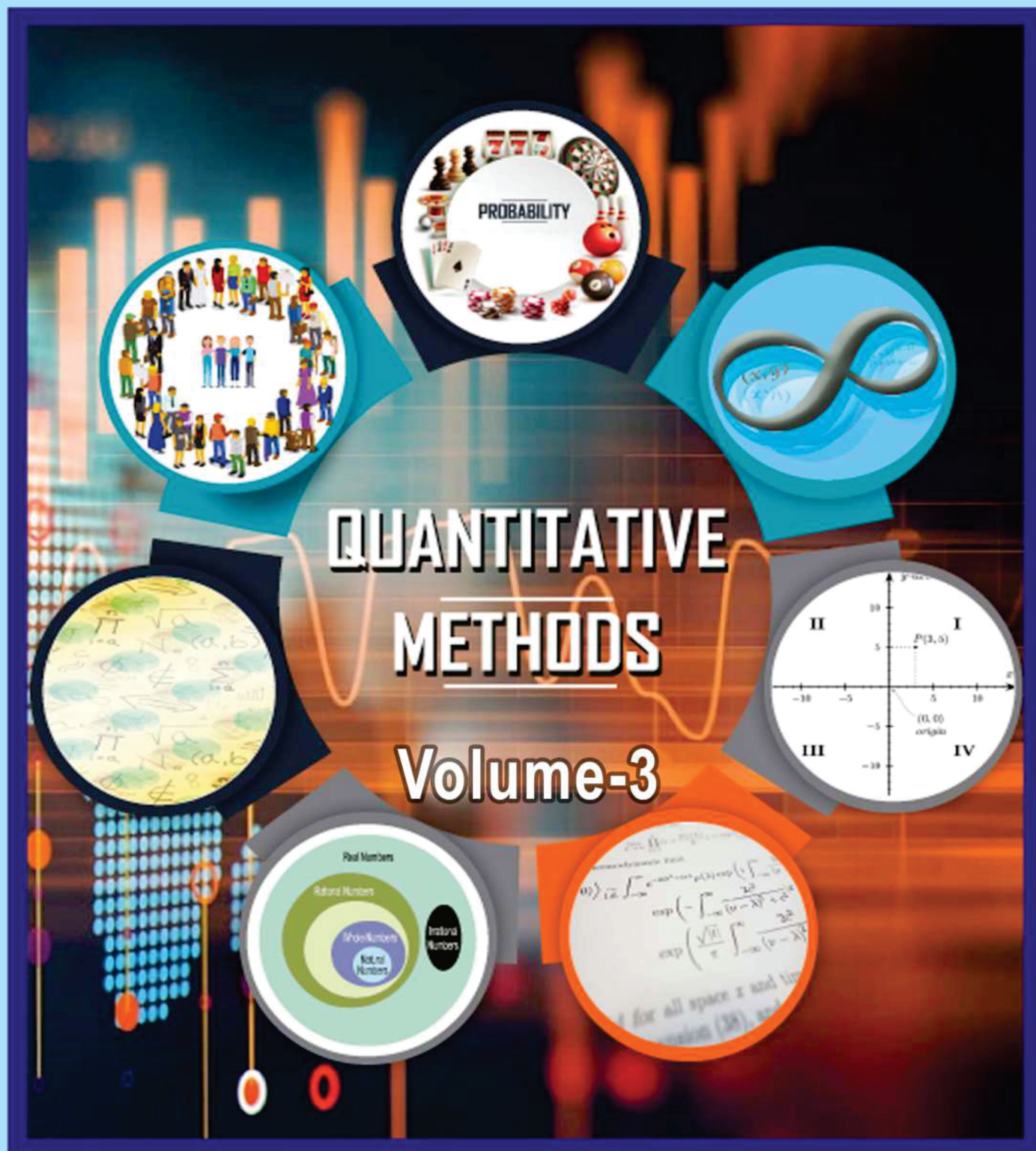


MEC-203

QUANTITATIVE METHODS



Volume-3

QUANTITATIVE METHODS

(Vol. 3)

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COURSE PREPARATION TEAM

Unit	Block/Unit Title	Unit Writer
BLOCK 1	REVIEW OF BASIC MATHEMATICS	
Unit 1	Fundamental Concepts of Mathematics	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
Unit 2	Overview of Basic Methods of Mathematics	
Unit 3	Relations and Functions	
Unit 4	Co-ordinate Geometry and Representation of Functions	Prof. Gopinath Pradhan
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Unit 5	Matrix Algebra	Sunando Basu
Unit 6	Vector Analysis	
Unit 7	Vectors and Matrices	
Unit 8	Vector and Matrix Representations of Linear Equations	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
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Unit 11	Differential Calculus: Functions of Several Variables	
Unit 12	Integration: Introduction and Techniques	Prof. Gopinath Pradhan
BLOCK 4	REAL ANALYSIS	
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Unit 14	Calculus of Several Variables	
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Unit 17	Unconstrained and Constrained Optimisation	Rittwik Chatterjee
Unit 18	Advanced Topics in Optimisation– I	Dr. Ram Prasad Yadav, Assistant Professor, Department of Mathematics, SRM University, Sonepat, Haryana
Unit 19	Advanced Topics in Optimisation – II	

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Unit 23	Intertemporal Optimisation-II	
Unit 24	Economic Applications of Dynamic Optimisation	
BLOCK 8	PROBABILITY AND PROBABILITY DISTRIBUTIONS	
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Unit 29	Sampling Distributions	
Unit 30	Estimation	Biswadeep Basu (upadation by Ms. Nivedita Mullick)
Unit 31	Hypothesis Testing	Biswadeep Basu

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MATHEMATICAL SYMBOLS/ NOTATIONS USED IN THE COURSE

Symbol	Symbol Name/Meaning Attached
$[a,b]$	Closed interval
(a,b)	Open interval
$\in$	Element of
Δ	Delta/Change in
dx	Differential
$\lim_{\Delta x \rightarrow 0} f(x)$	Limit
$\ln$	Natural log
$\int_a^b f(x) dx$	Definite Integral
$\forall$	Universal quantifier: "for all"
$\exists$	Existential quantifier: "there exists"
$\exists_1$	Existential quantifier: "there exists exactly one"
$\exists_{>1}$	Existential quantifier: "there exists more than one"
$\neg$	Negation; $\neg P$ "not P"
$\wedge$	Conjunction: "and"
$\vee$	Disjunction: "or"
$\Rightarrow$	Implication; $p \Rightarrow q$ "if p then q"
$\Leftrightarrow$	Equivalence: "if and only if"
$a, x \in \mathbb{R}$	Numbers, scalars
$a = (a_1, a_2, \dots, a_n) \in \mathbb{R}_+^n$	Vector of parameters
$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n$	Vector of variables
A, X	Sets
A, X	Matrices (with dimensions m by n) with m rows and n columns
$x \sim y$	Vector x is indifferent with respect to vector y
$\succ$	Strong preference relation
$x \succ y$	Vector x is strongly preferred over vector y
$\succsim$	(weak) Preference relation
$x \succsim y$	Vector x is (weakly) preferred over vector y
$\mathbb{R}$	Set of real numbers
$\mathbb{R}_+$	Set of nonnegative real numbers
$\text{int} \mathbb{R}_+$	Set of positive real numbers
$\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$	n – Dimensional space of real numbers, Cartesian product of set $\mathbb{R}$
$\mathbb{R}_+^n = \{x \in \mathbb{R}^n \mid x \geq 0\} \subset \mathbb{R}^n$	Nonnegative orthant (subspace) of set $\mathbb{R}^n$
$\text{int} \mathbb{R}_+^n \subset \mathbb{R}_+^n$	Interior of set $\mathbb{R}^n$
$f: X \rightarrow Y$	Function, X – Domain of a function (set of arguments of a function)

	Y – Codomain of a function (set of values of a function)
$y = f(x)$	Scalar function of one variable x – Independent variable (argument/input of a function) y – Independent variable (value of function)
$y = f(x_1, x_2)$	Scalar function of two variables x_1, x_2 – Independent variables (arguments/inputs of a function) y – Independent variable (value of function)
$\frac{dy}{dx}, y', f'(x)$	Derivative of first order of function $y = f(x)$
$\frac{d^2y}{dx^2}, y'', f''(x)$	Derivative of second order of function $y = f(x)$
$\frac{\partial y}{\partial x_i}, \frac{\partial f(x_1, x_2)}{\partial x_i}$	Partial derivatives of first order for function $y = f(x_1, x_2)$ with respect to variable x_i $i = 1, 2$
$\frac{\partial^2 f(x_1, x_2)}{\partial x_i \partial x_j}, \frac{\partial^2 f(x_1, x_2)}{\partial x_i^2}, i = 1, 2$	Partial derivatives of second order for function $y = f(x_1, x_2)$
$H(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f(x_1, x_2)}{\partial x_1^2} & \frac{\partial^2 f(x_1, x_2)}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(x_1, x_2)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x_1, x_2)}{\partial x_2^2} \end{bmatrix}$	Hessian, symmetric matrix of partial derivatives of 2 nd order for function $y = f(x_1, x_2)$
$\left. \frac{df(x)}{dx} \right _{x=\bar{x}}$	Value of a derivative of one variable function at point $\bar{x}$
$\left. \frac{\partial f(x_1, x_2)}{\partial x_i} \right _{\mathbf{x}=\bar{\mathbf{x}}}$	Value of a derivative of one variable function at point $\bar{\mathbf{x}} = (x_1, x_2)$
$\langle \mathbf{p}, \mathbf{x} \rangle = \sum_{i=1}^2 p_i x_i$	Scalar product of two vectors $\mathbf{p}, \mathbf{x} \in \mathbb{R}_+^2$
$\det(\mathbf{A})$ or $ \mathbf{A} $	Determinant of matrix $\mathbf{A}$
$d_E(\mathbf{x}^1, \mathbf{x}^2) = \left(\sum_{i=1}^2 (x_i^1 - x_i^2)^2 \right)^{\frac{1}{2}}$	Euclidean metric (distance)
$d(\mathbf{x}^1, \mathbf{x}^2) = \max_{i=1,2} \{x_i^1 - x_i^2\}$	Non-Euclidean metric (distance)
$\ \mathbf{x}_E\ = \left(\sum_{i=1}^2 (x_i)^2 \right)^{\frac{1}{2}}$	Euclidean norm
$\ \mathbf{x}\ = \max_{i=1,2} \{x_i\}$	Non-Euclidean norm
$t = 0, 1, 2, \dots$	Time as discrete variable
$t \in [0; +\infty)$	Time as continuous variable
$\nabla f, \text{grad} f, \text{Gradient of function } f, \nabla f = (\partial f / \partial x_1, \dots, \partial f / \partial x_n)$	

**SOME LETTERS OF THE GREEK ALPHABET (AND
CORRESPONDING SYMBOLS) YOU MAY ENCOUNTER**

Symbol	Greek Alphabet	Symbol	Greek Alphabet
α	Alpha	θ	Theta
β	Beta	λ	Lamba
γ	Gamma	π	Pi
δ	Delta	σ	Sigma
ε	Epsilon	χ	Chi
ψ	Psi	μ	Mu
ρ	Rho	ω	Omega

BLOCK 7

DYNAMIC OPTIMISATION

Unit 22	Intertemporal Optimisation-I
Unit 23	Intertemporal Optimisation-II
Unit 24	Economic Applications of Dynamic Optimisation
